Class 10 SST - Chapter 10: Minerals and Energy Resources

🌟 Introduction to the Chapter

The earth's crust is made up of different minerals embedded in rocks. Various metals are extracted from these minerals exclusively after proper refinement. Almost everything we use, from a tiny pin to a towering building or a big ship, is made from minerals.

📌 Classification of Minerals

Minerals can be broadly classified into:

1. **Metallic Minerals:** These contain metals. They are further divided into:
   - *Ferrous:* containing iron (e.g., Iron ore, Manganese, Nickel, Cobalt).
   - *Non-Ferrous:* not containing iron (e.g., Copper, Lead, Tin, Bauxite).
   - *Precious:* highly valuable (e.g., Gold, Silver, Platinum).
2. **Non-Metallic Minerals:** Do not contain metals (e.g., Mica, Salt, Potash, Sulphur, Granite, Limestone, Marble, Sandstone).
3. **Energy Minerals:** Used to generate power (e.g., Coal, Petroleum, Natural Gas).

Mode of Occurrence of Minerals

Minerals are usually found in 'ores' (an accumulation of any mineral mixed with other elements).

- In igneous and metamorphic rocks, minerals may occur in the cracks, crevices, faults, or joints.
- In sedimentary rocks, a number of minerals occur in beds or layers (they are formed as a result of deposition).
- Placer deposits: Certain minerals may occur as alluvial deposits in sands of valley floors.

⚡ Important Metallic Minerals: Iron Ore

Iron ore is the basic mineral and the backbone of industrial development.

- **Magnetite:** The finest iron ore with a very high content of iron up to 70%. It has excellent magnetic qualities.
- **Hematite:** The most important industrial iron ore in terms of the quantity used, but has a slightly lower iron content than magnetite (50-60%).
- *Major Iron Ore Belts in India:* Odisha-Jharkhand belt, Durg-Bastar-Chandrapur belt, Bellary-Chitradurga-Chikmagalur-Tumkur belt, and Maharashtra-Goa belt.

❌ Common Mistakes to Avoid

> [!WARNING]
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> 1. **Ores vs Minerals:** A mineral is naturally occurring inorganic substance with a definite chemical composition. An ore is a rock containing a high enough concentration of a specific mineral that it can be extracted profitably.
> 2. **Conventional vs Non-Conventional Energy:** Coal, oil, and gas are conventional (they have been used for a long time and will run out). Solar, wind, and tidal are non-conventional (they are relatively new and renewable).

📝 Quick Revision

Energy Resources

Energy is required for all activities. It is needed to cook, provide light and heat, propel vehicles, and drive machinery in industries.

- **Coal:** India is highly dependent on coal for meeting its commercial energy requirements. Types include Peat (low carbon), Lignite (low grade brown coal), Bituminous (most popular commercial coal), and Anthracite (highest quality hard coal).
- **Petroleum:** Provides fuel for heat and lighting, lubricants for machinery, and raw materials for many manufacturing industries.
- **Natural Gas:** Considered an environment-friendly fuel because of low carbon dioxide emissions. Used widely in industries and vehicles (CNG).
- **Non-Conventional Energy:** To reduce dependence on fossil fuels, India is heavily investing in Solar energy, Wind power, Biogas, and Geothermal energy.